

Exact Paper Faithful Reproduction Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the first coordinated `RCIM Model-Bank Reproduction` paper-faithful reproduction campaign prepared in:

- `doc/reports/campaign_plans/track_1/exact_paper/2026-04-10-21-47-55_exact_paper_faithful_reproduction_campaign_plan_report.md`

The package executed `4` repository-owned validation runs through the dedicated paper-faithful launcher:

- completed runs: `4`
- failed runs: `0`
- execution window: `2026-04-10T22:49:06.1087937+02:00` to `2026-04-11T20:08:29.5330140+02:00`
- campaign artifact root:
 - `output/training_campaigns/track1/exact_paper/forward/baseline_reproduction/shared/exact_paper_faithful_reproduction_campaign_2026_04_10_21_47_55/` Unlike a single-runner paper benchmark, this package used a coordinated structure:
- `2` exact-paper family-bank support runs for recovered-training fidelity and ONNX surface validation;
- `2` harmonic-wise shared-evaluator runs for the TE-level offline benchmark tied to `Target A` .

Objective And Outcome

The campaign had four concrete questions

1. can the recovered exact-paper family-bank branch still run stably under the current stack?
2. can the amplitude/phase exact-paper handoff be serialized cleanly even when ONNX export is disabled?
3. can dominant-harmonic specialization improve the shared offline evaluator tied to `Target A` ?
4. under one comparable offline evaluator, does the exact-paper-faithful direction beat the current repository baseline?

Outcome:

- both exact-paper support runs completed successfully and the recovered strict-support run exported `200` ONNX files with `0` failures;
- the harmonic-wise branch is now stable under the current Windows stack after threadpool hardening for `HistGradientBoostingRegressor` ;
- the specialized exact-paper dominant-harmonic run completed, but did not beat the current shared-evaluator baseline;
- the canonical campaign winner is the shared-evaluator reference run `track1_current_best_shared_evaluator_reference` ;
- `Target A` remains `not_yet_met` because the best test mean percentage error is `8.877%` versus the paper threshold `4.7%` .

Ranking Policy

This campaign contains two metric domains, so the ranking policy uses an explicit eligibility gate.

- eligible winner set: runs that emit the shared offline evaluator tied to Target A ;
- primary metric: test_mean_percentage_error_pct ;
- first tie breaker: test_curve_mae_deg ;
- second tie breaker: test_curve_rmse_deg ;
- third tie breaker: lexical run_name ;
- support-track note: exact-paper family-bank runs are serialized in the leaderboard, but they are not winner-eligible because they do not emit the TE-level shared evaluator metric.

Comparable Offline Ranking

Rank	Config	Target A	Test MPE [%]	Curve MAE [deg]	Curve RMSE [deg]
1	track1_current_best_shared_evaluator_reference	not_yet_met	8.877	0.002613	0.002812
2	exact_paper_dominant_harmonic_specialized	not_yet_met	9.123	0.002690	0.002885

Config	Oracle MPE [%]	Robot TE RMS [deg]	Cycloidal TE RMS [deg]
track1_current_best_shared_evaluator_reference	2.749	0.041350	0.032767
exact_paper_dominant_harmonic_specialized	2.749	0.041317	0.032944

Exact-Paper Support Runs

Rank	Config	ONNX Export	Winner	Mean Component MAPE [%]	Mean Component MAE
1	exact_paper_recovered_reference	yes	RF	18.369	0.056284
2	exact_paper_amplitude_phase_reference	no	RF	18.369	0.056284

Config	Mean Component RMSE	ONNX Exported	Failed Exports
exact_paper_recovered_reference	0.144839	200	0
exact_paper_amplitude_phase_reference	0.144839	0	0

Campaign Winner

The explicit campaign winner is

- `track1_current_best_shared_evaluator_reference`

Its result was:

- test mean percentage error: `8.877%`
- test curve MAE: `0.002613 deg`
- test curve RMSE: `0.002812 deg`
- oracle test mean percentage error: `2.749%`
- Target A threshold: `4.7%`
- Target A status: `not_yet_met`

This run won because it is the lowest-error candidate inside the only metric domain that is directly comparable to the paper-facing offline benchmark used by this campaign.

Interpretation By Experiment Block

1. The Exact-Paper Support Surface Is Now Operational

The exact-paper branch completed in both prepared forms during this campaign:

- `exact_paper_recovered_reference` kept the recovered family bank stable and exported `200` ONNX files with `0` failures;
- `exact_paper_amplitude_phase_reference` completed cleanly with export disabled, which now exercises the same report path without schema regressions.

Interpretation:

- the exact-paper branch is no longer blocked by the prior `HGBM` export issue or by the disabled-export report-builder bug;
- these runs now act as reusable support evidence for paper-faithful parameterization and deployment-facing artifact preparation.

2. The Shared Evaluator Still Favors The Current Repository Baseline

The comparable harmonic-wise ranking stayed in the same order as the pre-campaign repository state:

- `track1_current_best_shared_evaluator_reference` : `8.877%`
- `exact_paper_dominant_harmonic_specialized` : `9.123%`

Interpretation:

- the exact-paper-inspired specialization remains viable, but it did not beat the current best shared-evaluator reference;
- the gap is small enough to remain interesting, but the campaign does not justify promoting the specialized branch over the current baseline.

3. Target A Is Still Open

The campaign moved the exact-paper reproduction branch from fragile to runnable, but it did not close the paper threshold:

- best achieved test mean percentage error: 8.877%
- paper threshold: 4.7%
- verdict: not_yet_met

Interpretation:

- this campaign improves reproducibility and comparability more than raw offline accuracy;
- the next RCIM Model-Bank Reproduction improvement cycle should focus on reducing TE-level curve error, not on more family-bank stabilization work.

4. Playback Remains Repository-Owned And Comparable

The winner keeps the playback path alive for both canonical offline styles:

- robot mean reconstructed TE RMS: 0.041350 deg
- cycloidal mean reconstructed TE RMS: 0.032767 deg

Interpretation:

- the campaign keeps the repository aligned with the paper-facing offline evaluation shape;
- however, these playback summaries still sit inside an offline-only comparison scope and do not yet constitute the paper online benchmark.

Recommended Next Step

The correct next step is not another exact-paper export campaign. It is a new RCIM Model-Bank Reproduction offline-improvement iteration that starts from the shared-evaluator winner and attacks the remaining TE-level error directly, while reusing the now-stable exact-paper support surface as evidence and artifact infrastructure.